Predicting Imminent Errors from Brain Activity

Using Pre-Response EEG Signals in the Eriksen Flanker Task

Manya Yadavilli · Enrique Aranda · Shien Tioh · Travis Huang · Eric Nguyen
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Abstract

Can the brain predict its own mistakes? In this project, we investigated whether EEG signals in the 500 ms before a response can predict upcoming errors. Participants played the Eriksen Flanker task, a fast reaction-time task that reliably produces errors, while EEG data were recorded. We then trained an SVM classifier to distinguish between trials that would end in an error versus correct responses using only pre-response brain activity.

We focused on three frequency bands linked to cognitive control and performance: frontal theta (4–7 Hz), posterior alpha (8–12 Hz), and central beta (13–30 Hz). EEG data were recorded from 8 electrodes (Fz, F3, Cz, C3, C4, Pz, O1, O2) and preprocessed using a 40Hz lowpass, linear detrending, and baseline correction. Band power features were extracted and used to train an SVM classifier. Results will evaluate whether pre-response neural activity contains enough information to predict errors above chance.

1. Introduction

Most people have experienced the feeling of realizing they made a mistake right after responding. In neuroscience, this is associated with the Error-Related Negativity (ERN), a well-known EEG signal that occurs shortly after an error is made. However, a more interesting question for brain-computer interfaces (BCIs) is whether the brain is already in an “error-prone” state before the mistake occurs.

This project asks: **can EEG signals in the 500 ms before a response predict whether that response will be correct or incorrect?** If so, this suggests that errors are not random, but instead arise from measurable neural states. This has potential applications in real-time error prevention in areas like medicine, aviation, and rehabilitation.

The Eriksen Flanker task is well-suited for this question because it reliably induces response conflict and errors. It is also known to produce well-characterized EEG signals such as the N2 and ERN, making it a strong paradigm for studying performance monitoring.

In this project, we recorded EEG from two participants, extracted spectral features from a pre-response window, and trained classifiers to distinguish correct versus error trials. We used the 8-channel Open-BCI setup that was provided to us.

2. Background

2.1 The Eriksen Flanker Task

The Eriksen Flanker task (Eriksen & Eriksen, 1974) requires participants to respond to a central target stimulus while ignoring surrounding distractors. On congruent trials (→→→→→), all arrows point in the same direction, while on incongruent trials (←←→←←), the flankers conflict with the target. Incongruent trials create response conflict, leading to slower reaction times and increased errors.

This task is widely used because it engages brain regions involved in conflict monitoring and error processing, particularly the anterior cingulate cortex (ACC).

2.2 EEG Components in Error Processing

- **Error-Related Negativity (ERN):**
  The ERN is a negative deflection occurring ~50–150 ms after an error, typically observed at fronto-central sites (e.g., Fz/Cz). It reflects rapid internal error detection (Gehring et al., 1993).
- **N2 Component:**
  The N2 peaks ~200–350 ms after stimulus onset and is larger during conflict (incongruent trials). It is associated with conflict detection processes (Nee et al., 2014).
- **Error Positivity (Pe):**
  The Pe is a later positive component linked to conscious awareness of errors. While important, it is not the focus of this project since we are interested in pre-response activity.
- **N2 (conflict detection) → PMP (post-monitoring process) change → response → ERN → Pe.**

2.3 Pre-Error Oscillatory Signatures

In addition to ERPs, oscillatory activity before a response may indicate whether an error is likely.

- **Theta (4–7 Hz, frontal):**
- Associated with cognitive control and conflict monitoring. Increased frontal theta has been linked to performance monitoring and control engagement (Cavanagh & Frank, 2014).

- **Alpha (8–12 Hz, posterior):**
- Linked to attention. Higher alpha is often associated with reduced visual processing or attentional disengagement (Kelly et al., 2009).

- **Beta (13–30 Hz, central):**
- Related to motor preparation. A decrease in beta power typically reflects movement preparation, and reduced desynchronization may indicate weaker motor readiness (Pfurtscheller & Lopes da Silva, 1999).

3. Research Question & Hypotheses

Our central question: Can pre-response EEG features (−500 to 0 ms before button press) reliably predict whether the upcoming response will be an error or correct?

Hypothesis: Based on the literature, we predict the following in the pre-response window on error vs. correct trials:

Band	Direction	Electrodes	What we think it means
Theta (4–7 Hz)	↑ Increase	Fz, F3, Cz	Frontal control system not engaging strongly enough - error-prone state
Alpha (8–12 Hz)	↑ Increase	Pz, O1, O2	Visual attention lapsing - participant mentally drifting before the response
Beta (13–30 Hz)	↓ Decrease	Cz, C3, C4	Insufficient motor prep - the movement is executed without full readiness

If classifiers trained on these features perform above chance, that confirms errors emerge from a distinct neural state rather than random noise.

4. Methods

4.1 Participants

Two participants completed the task. All had normal or corrected-to-normal vision and no reported neurological conditions. Running ourselves as subjects had the practical advantage of full flexibility for data collection sessions, though it does mean we need to be careful about any potential bias in data handling. Each person provided informed consent to participate.

4.2 Task Design

We implemented the Eriksen Flanker task in PsychoPy. Each trial presented five arrows for 200 ms, followed by a jittered blank screen (800–1200 ms). Participants responded using F/J keys based on the central arrow direction, F for left and J for right. Trials were ~50% congruent and ~50% incongruent, with 200 trials for one participant and 600 for the other.

4.3 EEG Recording

EEG was recorded from 8 electrodes following the 10-20 international system. We chose these 8 based specifically on the bands we wanted to capture:

#	Electrode	Band Focus	Why here?
1	Fz	Theta	Midline frontal - main ACC theta site
2	F3	Theta / Beta	Left frontal, N2 signal for right-hand responses
3	Cz	Theta / Beta	Midline central - motor and control overlap
4	C3	Beta	Left central - right-hand motor cortex
5	C4	Beta	Right central - bilateral motor coverage
6	Pz	Alpha	Midline parietal - attentional alpha
7	O1	Alpha	Left occipital - visual alpha maxima
8	O2	Alpha	Right occipital - bilateral visual coverage

Impedances were kept below 100kΩ. Ground and reference electrodes included.

4.4 Preprocessing

EEG data were preprocessed as follows:

1. 40 Hz Lowpass filtering
2. Linear detrending
3. Epoching: −500 to +400 ms around response
4. Baseline correction: −500 to −400 ms
5. Artifact rejection: trials exceeding ±150 μV were removed

To assess data quality, we analyzed the number of trials removed during artifact rejection using a ±150 µV threshold. Due to high-amplitude noise, particularly in one participant's dataset, we had to exclude a substantial amount of data. This suggests that some recordings were noisier than expected, which may have reduced the number of usable trials for classification. (Figure 1)

We also observed that certain channels (especially C4) consistently showed higher noise levels compared to others, which may reflect electrode placement issues or poor contact during recording. Because error trials are already relatively rare, removing noisy trials further reduced the number of available error examples..

4.5 Feature Extraction

The feature window was **−500 to 0 ms before response**.

We extracted:

- **Band power (Welch's method)**

- Theta (4–7 Hz)

- Alpha (8–12 Hz)

- Beta (13–30 Hz)
  → across all 8 electrodes (**24 features per trial**)

(Additional analyses such as ERPs were used for validation but not as classifier features.)

4.6 Classification

We trained an SVM classifier with an RBF kernel using the 24 spectral features. To address the class imbalance, we used the stratified 5-fold cross-validation so that each fold maintained the ratio of correct to error trials. Model performance was evaluated using AUC-ROC, together with sensitivity and specificity. To understand which features contributed most to our model, we used permutation-based feature importance, measuring the mean AUC decrease when each feature was shuffled.

5. Results

5.1 EEG Signals

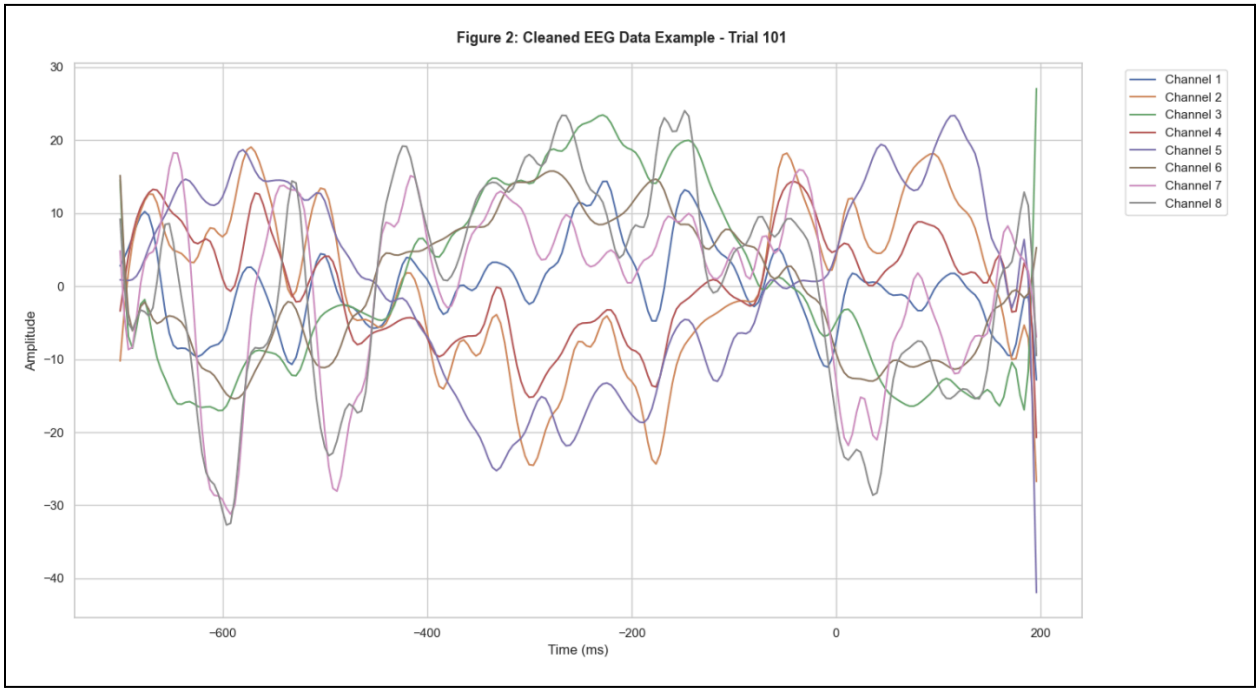
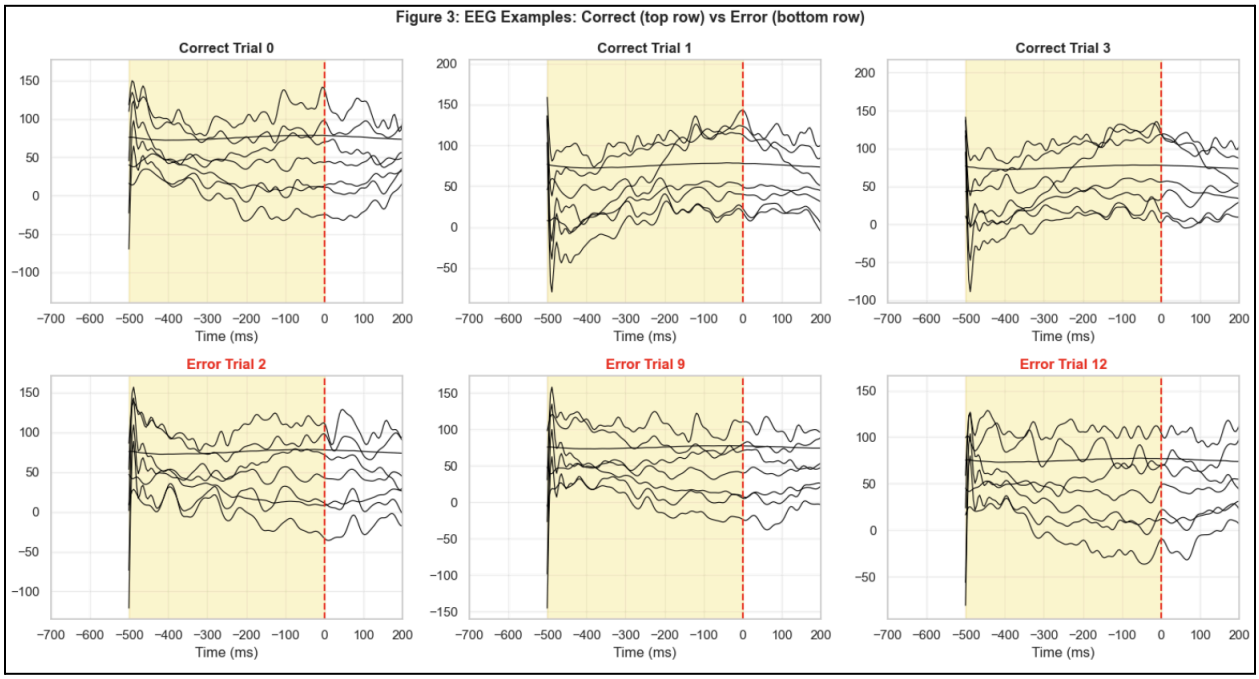


Figure 2 demonstrates what the Clean EEG data looks like after our lowpass filtering and removal of artifacts over the threshold of 150 microvolts. We view trial 101 as it provides a quick example of how the channels look around the time of the button press. Our Flanker stimulus script recorded from 700 ms before the button press to 200 ms after so we see the signals completely stop at 200 ms exactly.



Here, figure 3 demonstrates the 8 channels within the time frame that we're focusing on (500 ms before button press to 0ms). We put the correct trials adjacent to the error trials to see if we can see any differences in the signals between the two. The markers at 0 ms also give us a clearer view at what happens to the waves right after button press. Based on the three examples here, we see the correct trials are more stable after button press, whereas the incorrect trials tend to jolt a little more after button press. Due to non-stationarity, it's hard to compare these individual trials side by side.

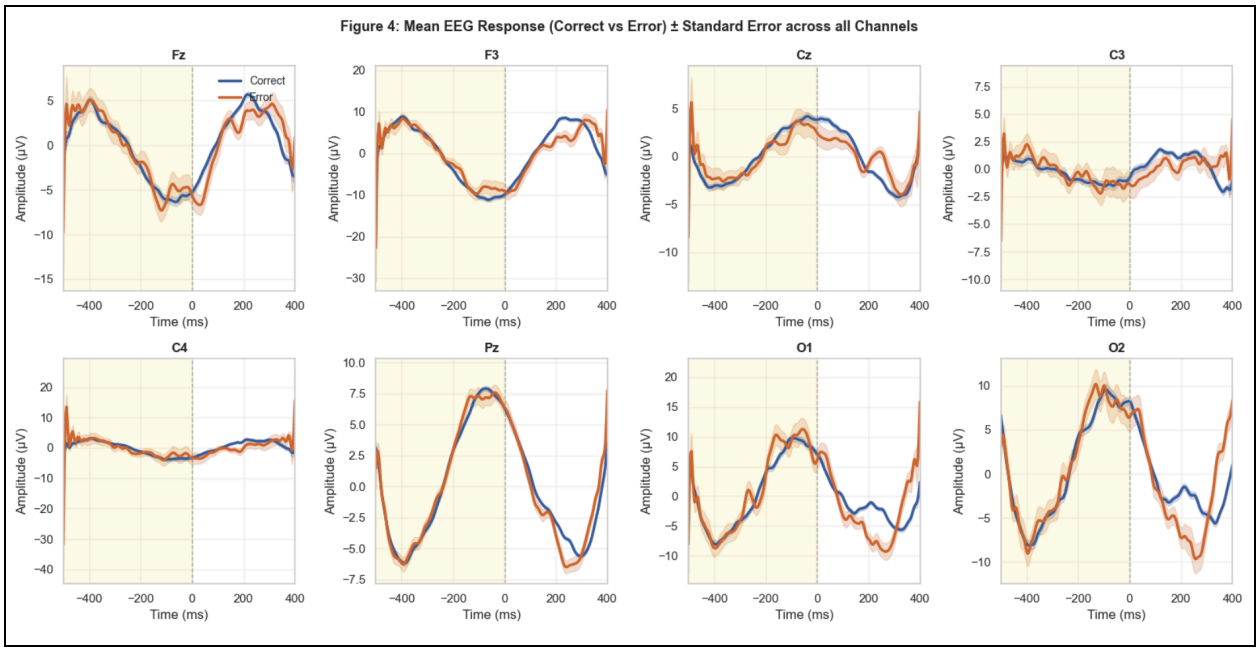


Figure 4 overlays the mean incorrect trials over the mean correct trials across all channels to be able to see the difference more clearly. We can see in the O1 and O2 channels, there's a larger difference in the signal prior to button press. This could possibly be due to the incongruity trials where the Flanker in the center is adjacent to opposite pointing arrows (>><>>). We see around 400 ms before the button that there's a jump in O1 and O2, but also in Pz and Cz, that suggests when the stimulus is presented. Due to the few errors in our data, the sampling size between the two means can not make for a fair comparison.

5.2 Classifier Performance

The SVM classifier trained on 24 features (theta, alpha, and beta band power across 8 channels) achieved an AUC of 0.489 under 5-fold cross-validation, which is essentially at chance level (AUC = 0.50). Figure 5.1 is a confusion matrix that reveals that the classifier predominantly predicted the majority class (correct trials), correctly labeling 416 correct trials but only 22 error trials out of 59 total. This pattern suggests that there weren't enough error trials to fully be capable of predicting errors. Figure 5.2 shows us that our SVM did not learn a meaningful boundary between correct and incorrect, therefore not being able to predict any better than chance. These figures show us that an unequal distribution of correct and error trials prove to be challenging for our SVM classifier.

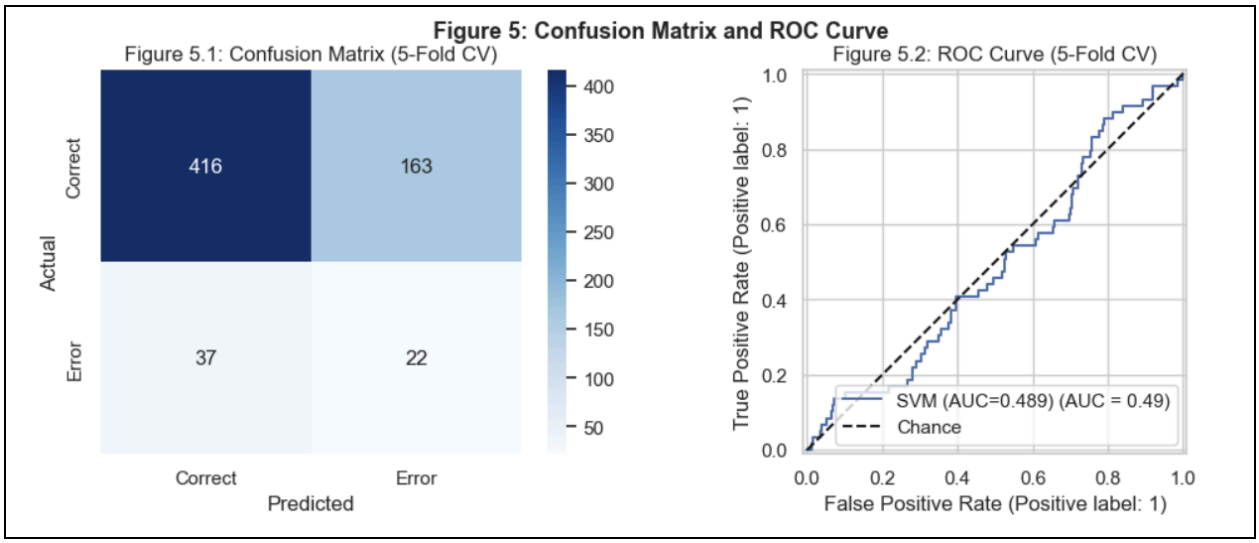


Figure 5 demonstrates the classifier performance under 5-fold cross validation. On the left is a confusion matrix showing predicted vs. actual trial labels. The classifier predominantly predicted correct responses. It correctly identifies 416 correct trials but only 22 of 59 total error trials, and this indicates the strong class imbalance. On the right is a ROC curve for the SVM classifier (blue line) vs. chance-level performance (dashed line). The AUC of 0.489 indicates performance at chance, which suggests that pre-response spectral features by themselves were not enough to distinguish errors from correct responses.

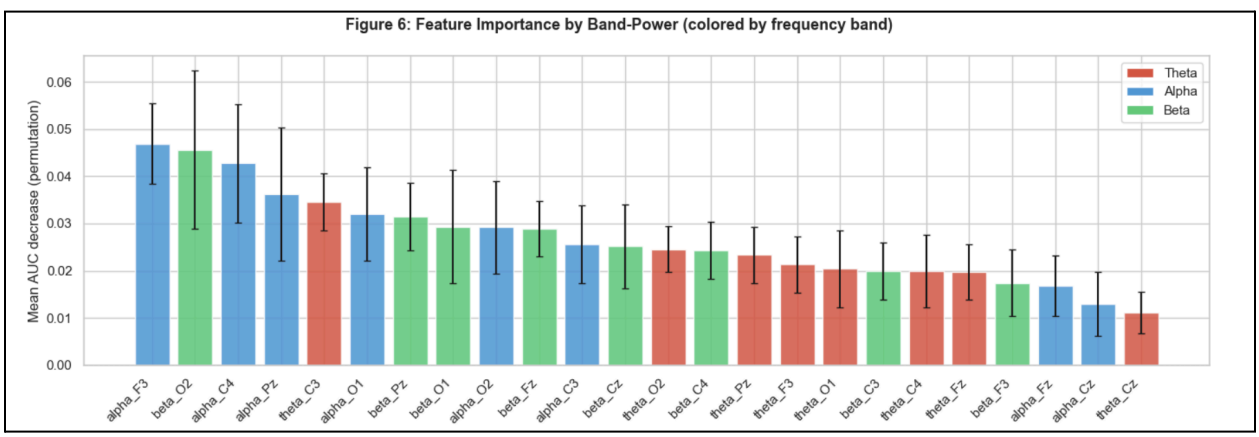
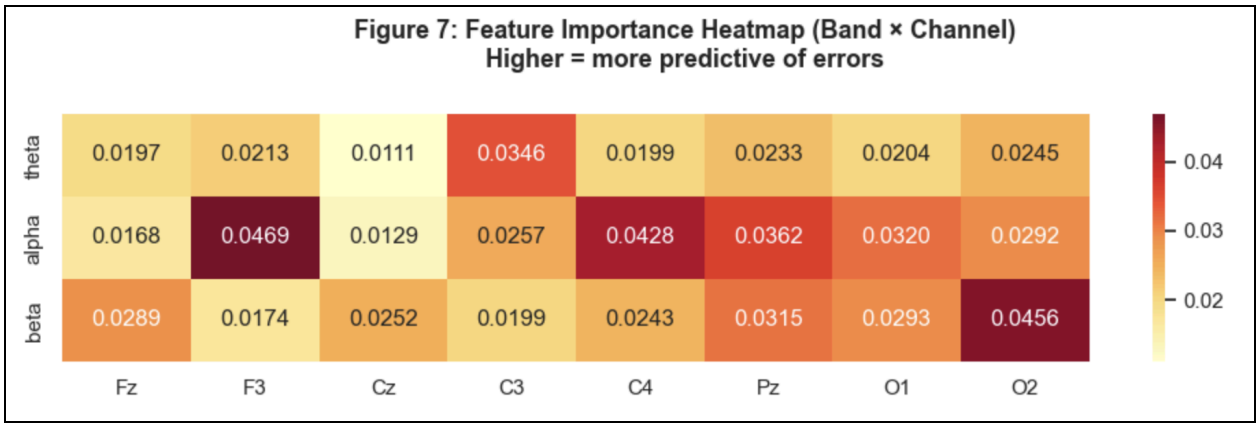


Figure 6 shows the permutation-based feature importance ranked by mean AUC decrease. Each bar represents one of the 24 features (3 frequency bands x 8 channels) color-coded by bands. Error bars represent standard deviation across permutation folds. Alpha power at F3 and beta power at O2 showed the highest feature importance, which indicates that they have a greater contribution to classifier predictions. On the other hand, theta at Cz showed the lowest feature importance.



Lastly, figure 7 shows the heatmap of permutation feature importance organized by frequency band (represented by rows: theta, alpha, and beta) and electrode (represented by columns: Fz, F3, Cz, C3, C4, Pz, O1, and O2). Warmer colors indicate higher predictive importance and lighter colors indicate lower predictive importance. The highest importance features were alpha at F3 (0.0469) and beta at O2 (0.0456), while the lowest importance feature was theta at Cz (0.0111).

6. Discussion

6.1 What We Expected to See (and Why)

If our hypotheses are supported, it would imply that errors in the Flanker task aren't just random. Instead, they may happen when the brain is in a kind of "off" state. This could mean weaker cognitive control (higher frontal theta), lower attention (higher posterior alpha), and less motor preparation (reduced central beta). This idea lines up with previous work (van Driel et al., 2012), which suggests that different types of errors can come from things like attention lapses or control failures.

The classifier performance will be the main way we test this. Even moderate results (like AUC around 0.62–0.70) would still be meaningful, since predicting errors from single-trial EEG before a response is pretty challenging. For example, Kato et al. (2015) reported around 62% accuracy using a similar setup, while DeLaRosa et al. (2020) reported higher accuracy, but they used more electrodes and included post-stimulus features, which makes the task easier.

6.2 Limitations

- Small N: Two participants isn't enough to make strong generalizations. Individual differences in scalp topography and EEG noise levels are real, and with two people, one noisy dataset can heavily affect the average.
- Limited channels: 8 electrodes gives us targeted coverage but no spatial resolution for source localization. We can say what band at what scalp location, not what brain structure.

- Our Flanker stimulus script attempted to filter after epoching which we had to correct through linear detrending
- We had an amplitude scaling inconsistency issue where a majority of our first subject's trials had to be dropped as they were over the threshold of 150 microvolts.
- Class imbalance: There will always be fewer error trials than correct trials. Even with stratified cross-validation, this can affect classifier calibration, especially for sensitivity estimates.
- Pre-response window contamination: The −500 to 0 ms window may include post-stimulus processing rather than purely preparatory activity. Without controlling for RT differences between error and correct trials, some of the pre-response signal could actually be late sensory processing rather than motor preparation.
- ICA limitations with 8 channels: ICA quality scales with electrode count. With only 8 channels, we have less statistical leverage to cleanly separate neural from artifactual components, especially for subtle muscle artifacts.

6.3 Future Directions:

Future improvements could include:

- More participants
- More trials
- Higher-density EEG
- Real-time classification
- Testing deep learning models (e.g., EEGNet)

7. Conclusion

In this project, we investigated whether pre-response EEG signals could predict imminent errors in an Eriksen Flanker task. While our SVM classifier achieved chance-level performance (AUC = 0.489) due to a severe class imbalance and limited-data, our feature importance analysis revealed that alpha and beta power at frontal and occipital sites contributed most to the model's decision boundary. Ultimately, predicting single-trial errors in real-time requires more subjects to train on and potentially a game that would induce more errors, though we believe we paved the way for further research to expand upon.

References

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